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RESEARCH ARTICLE

First-Order Absorbing Markov Chain Analysis for Web Navigation and Conversion Rate Optimization

SHUNE LAE AUNG^{ID} AND GENDO KUMOI^{ID}, (Member, IEEE)

Nagaoka University of Technology, Nagaoka 940-2188, Japan

Corresponding author: Shune Lae Aung (s247011@stn.nagaokaut.ac.jp)

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ABSTRACT Improving conversion rates (CVR) in e-commerce and web-based online applications requires understanding and optimizing user navigation behavior. This study employs a first-order absorbing Markov chain (AMC) framework to model user journeys through websites, where transient states represent web pages, and absorbing states denote conversion or exit events. Unlike traditional attribution methods that retroactively assign credit to past touchpoints, our AMC framework enables both diagnostic analysis—quantifying each web page's contribution to overall conversion probability—and prescriptive simulation by evaluating hypothetical structural changes, such as modifications to link transition probabilities. Experiments on both synthetic datasets—representing diverse browsing patterns that include goal-oriented flows, random browsing, and leaky funnels—and real-world e-commerce clickstream data show that the AMC approach provides more interpretable and robust insights than a logistic regression baseline, particularly in low-CVR scenarios common in e-commerce environments. A key contribution is the introduction of a conversion impact metric that serves as a forward-looking decision-support tool by quantifying how individual link modifications influence overall conversion probability, enabling data-driven decision-making for website optimization. Through its emphasis on interpretability and transparency, this framework demonstrates the practical utility of AMC-based approaches for data-driven user experience design and website structure optimization.

INDEX TERMS First-order absorbing Markov chain, conversion rate optimization, customer journey, web navigation, web browsing behavior analysis.

I. INTRODUCTION

Analyzing user behavior on e-commerce platforms has become essential for improving conversion rates (CVR). A conversion—a desired user action such as a purchase or registration—is directly related to revenue and user engagement, making CVR optimization a critical focus for online businesses [1].

To better understand the user journey, researchers have employed various computational approaches. Advanced sequential models, such as recurrent neural networks (RNNs) [2], traditional machine learning methods [3], sequence pattern mining [4], and broader web mining

techniques [5], have all been used to identify drop-off points and improve navigation flows [6]. While these methods demonstrate strong predictive capability, they often exhibit limited interpretability, making it difficult for practitioners to understand which user behaviors most strongly influence conversions [7]. Moreover, when dealing with low conversion rates—a common condition in e-commerce—such models face challenges with data imbalance and unstable predictions [8]. This highlights the need for approaches that not only achieve predictive accuracy but also yield transparent and actionable behavioral insights.

Logistic regression (LR) remains a popular baseline due to its simplicity and interpretability. However, its assumption of independent observations [9] is not well-suited for sequential clickstream data, where each user action depends

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on prior steps. Consequently, LR cannot natively capture process-flow dynamics and often struggles under low-CVR conditions [10].

To address these limitations, researchers have increasingly turned to Markov Chain models, which represent user navigation as a sequence of probabilistic transitions between pages. For instance, Lee [11] modeled airline booking behavior using a stochastic Markov framework that quantified the contribution of each step in the booking funnel, offering actionable insights into user decision processes. Building upon this line of work, the present study models user navigation as a stochastic process in which web pages act as transient states and conversion or exit events as absorbing states. This absorbing Markov chain model framework enables the estimation of each page's contribution to overall conversion and supports "what-if" analyses that simulate the effect of link modifications.

Given the nature of website browsing behavior, this study employs a first-order AMC, in which the next page transition depends only on the current page. Prior research has shown that first-order Markov models can adequately capture user navigation dynamics, as actions taken two or more steps earlier typically exert little influence on immediate next-step behavior [12], [13], [14]. This balance between simplicity and behavioral fidelity preserves a high level of interpretability while avoiding the state explosion associated with higher-order models.

A primary contribution of this research is the introduction of a conversion impact metric that quantifies how specific link-level modifications influence the overall conversion probability. Although "what-if" analyses have been explored in prior Markov-based studies, this work uniquely applies such analysis to conversion rate optimization by examining the granular effect of modifying a single internal link, specifically by adjusting one transition probability in the website navigation graph. This fine-grained focus enables interpretable simulations that reveal how targeted design changes could alter user flow and ultimately increase conversion probability. Although multiple simultaneous modifications are possible, beginning with single-link adjustments ensures controlled analysis and clearer attribution, providing a foundation for future extensions.

This paper evaluates the effectiveness of the proposed AMC method in two areas: (1) estimating each web page's contribution under low-CVR conditions, and (2) evaluating the impact of hypothetical link modifications via the conversion impact metric. Experiments are conducted using both synthetic datasets, which allow controlled and systematic validation, and real-world clickstream data to confirm practical applicability. This combined evaluation framework strengthens both theoretical validity and empirical robustness.

The research addresses a critical gap in existing approaches: the lack of prescriptive and interpretable frameworks for granular website optimization, particularly in low-CVR environments where traditional statistical models struggle to capture sequential user dynamics. The proposed

framework models the website ecosystem as an Absorbing Markov Chain (AMC), consisting of transient states (web pages) and absorbing states (conversions or exits). Unlike traditional marketing attribution methods that focus on retroactive credit assignment to past touchpoints, the proposed conversion impact metric serves as a forward-looking decision-support tool. By quantifying how marginal changes in transition probabilities propagate through the fundamental matrix, the framework enables data-driven "what-if" simulations to predict the impact of structural changes on overall conversion probability prior to implementation. This work introduces the first AMC-based conversion impact metric for granular, prescriptive link-level optimization—bridging the gap between interpretable diagnostics and actionable structural recommendations in low-CVR e-commerce environments.

II. RELATED WORK

Our research builds upon a well-established body of work that uses stochastic processes to model sequential user behavior. This section first outlines the core mathematical concepts of first-order absorbing Markov chain that form the foundation of our model. It then reviews key applications of these models in user behavior modeling, marketing attribution, strategic simulation, and customer lifecycle analysis, situating our contributions within this broader context.

A. CORE CONCEPTS OF FIRST-ORDER ABSORBING MARKOV CHAINS

The foundational theory of AMCs, largely developed by Kemeny and Snell [15], provides a mathematical toolkit for analyzing systems that evolve through a sequence of states until reaching a terminal (absorbing) state. A Markov chain is defined as absorbing if it contains at least one absorbing state and if it is possible to reach an absorbing state from every transient (non-absorbing) state.

The dynamics of such a system are captured in the transition matrix P , canonically partitioned as:

$$P = \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix}$$

Here, Q describes transitions between transient states, R represents probabilities of moving from transient to absorbing states, I is the identity matrix for absorbing states, and 0 is a zero matrix indicating that once an absorbing state is reached, it cannot be left.

Two critical matrices are derived from this structure. The fundamental matrix $N = (I - Q)^{-1}$ gives the expected number of visits to transient states starting from a given state. The absorption probability matrix $B = NR$ provides the probability of ending in each absorbing state from any transient state [16].

While AMCs can be extended to higher-order chains or Hidden Markov Models (HMMs) [17], these often introduce significant complexity. Given the nature of website browsing behavior, first-order Markov models are generally

adequate—what a user viewed two or more steps back typically has little influence on the next page visited [12], [13], [14]. Prior work demonstrates that first-order models capture user navigation effectively while keeping state complexity manageable [12]. Higher-order models, though sometimes more accurate, can lead to exponential growth in states and computational cost [13]. Additionally, combining first-order transition probabilities with structural information, such as page ranking, can enhance predictive performance without sacrificing interpretability [14]. Thus, our study deliberately employs a first-order AMC to preserve interpretability and enable actionable, data-driven simulations.

B. APPLICATIONS IN USER BEHAVIOR MODELING

Markov chains have effectively proved in modeling user navigation on websites. Sarukkai [12] applied first-order Markov models to predict the next page a user would visit, achieving improved accuracy over baseline methods. Deshpande and Karypis [13] extended this by selectively incorporating higher-order information where predictive gains were substantial, balancing complexity with performance. Similarly, Zukerman et al. [17] explored Hidden Markov Models for web navigation, showing how latent states can capture browsing modes not directly observable.

Beyond prediction, Markov models have been used to understand user intent and engagement patterns. Dutta and Singh [14] integrated link structure and transition probabilities to recommend navigation paths that increase user satisfaction. These studies collectively highlight the interpretability and flexibility of Markov-based approaches—qualities that are central to our application in CVR optimization.

While advanced models such as Deep Reinforcement Learning (DRL) have been applied to navigation problems, interpreting the decision-making process for long-horizon tasks remains a significant challenge due to high computational complexity and a lack of model stability [18]. In contrast, the first-order AMC provides a transparent and noise-robust alternative that preserves behavioral fidelity while remaining computationally efficient for long-horizon user journeys.

C. MARKETING ATTRIBUTION AND CUSTOMER JOURNEYS

Marketing attribution seeks to assign credit to various touchpoints along a customer's path to conversion. Traditional heuristics (e.g., last-click attribution) oversimplify complex multi-step journeys. Markov chains offer a data-driven alternative: by modeling customer interactions as states and channel exposures as transitions, researchers can estimate each channel's marginal contribution to conversion [19].

Anderl et al. [19] demonstrated that Markov chain-based attribution models provide more equitable credit distribution across touchpoints than heuristic rules. Xu et al. [20] proposed probabilistic attribution models that account for causal relationships between exposures, leveraging

attribution modeling coefficients to quantify removal effects—how much conversion probability would drop if a specific touchpoint were removed.

Our research aligns with this literature by treating web pages as “touchpoints” and conversion as the terminal state. The conversion impact metric we introduce extends these attribution concepts to structural modifications, enabling “what-if” analyses that simulate the effect of changing link probabilities rather than removing touchpoints entirely.

D. STRATEGIC SIMULATION AND SCENARIO ANALYSIS

Markov models are well-suited for scenario simulation. In airline revenue management, Lee [11] used AMCs to simulate booking funnel behavior, quantifying how process changes—such as simplifying checkout steps—affect overall booking rates. This approach allowed decision-makers to evaluate interventions before implementation, reducing risk and cost.

Ching et al. [21] applied Markov models to customer relationship management, simulating how retention strategies alter customer lifecycle value. By adjusting transition probabilities between customer states (e.g., active, dormant, churned), they estimated the long-term revenue impact of marketing interventions.

Our framework adopts a similar simulation logic: by modifying a single transition probability in the web navigation graph, we can estimate the resulting change in overall conversion probability. This “what-if” capability provides actionable guidance for UX designers and website managers seeking to optimize navigation flows.

E. COMPARISON WITH MACHINE LEARNING AND DEEP LEARNING APPROACHES

While recent studies have applied neural networks to predict conversions [2] or next-page visits [6], such models often operate as closed-box models, offering limited insight into why specific pages, transitions, or navigation paths contribute to conversion outcomes. Interpretability is especially critical in conversion rate (CVR) optimization settings, where stakeholders must understand which concrete design changes to implement [7]. Logistic regression partially addresses interpretability through feature coefficients; however, it assumes independent observations, thereby violating the inherently sequential and path-dependent nature of clickstream data [9]. Prior studies comparing Markov-based journey models with logistic regression have shown that Markov approaches better capture process dynamics, particularly in environments characterized by low conversion rates and strong path dependencies [10].

In the current landscape of sequential modeling, deep learning architectures have set new benchmarks for predictive accuracy. Recurrent Neural Networks (RNNs), such as the GRU4Rec framework, utilize gated mechanisms to track evolving user states within a session [22]. Furthermore, transformer-based approaches like Alibaba's Behavior Sequence Transformer (BST) and BERT4Rec leverage

self-attention to capture long-range behavioral dependencies across vast clickstream datasets [23], [24].

However, the closed-box model nature of these complex models remains a major bottleneck for adoption in mission-critical e-commerce domains, where stakeholders require transparency to minimize false outcomes [25]. While deep learning excels at predicting the next click, our proposed AMC framework provides superior prescriptive utility. Unlike the latent vector spaces used in RNNs or Transformers, the AMC provides a direct mathematical mapping—via the fundamental matrix $N = (I - Q)^{-1}$ —that identifies exactly how a single link-level perturbation influences the global conversion probability. By utilizing an AMC framework, this study provides the necessary explainability to identify precisely how specific link-level changes influence conversion—offering granular insights that opaque models often fail to provide.

Similarly, although higher-order or variable-order Markov models (VOMMs) may yield incremental improvements in predictive accuracy, they incur substantial state-space expansion that obscures the direct relationship between individual transitions and conversion outcomes [26], [27]. By intentionally adopting a first-order, memoryless formulation, our framework enables transparent counterfactual “what-if” simulations that emphasize decision relevance over marginal gains in predictive performance [28], [29].

Building on these considerations, our comparative analysis presented in Section V evaluates AMC alongside logistic regression using both synthetic and real-world datasets. The results demonstrate that AMC provides more stable and interpretable estimates under low-CVR conditions, reinforcing its practical value as a decision-support framework for prescriptive website optimization.

F. RESEARCH GAP AND CONTRIBUTION

While existing work demonstrates the utility of Markov models in user behavior prediction, attribution, and simulation, few studies focus specifically on using AMCs for granular link-level optimization in low-CVR e-commerce settings. Our contribution lies in:

- 1) Introducing a conversion impact metric that quantifies the effect of single-link modifications on overall conversion probability.
- 2) Demonstrating the framework’s robustness, uncertainty-aware evaluation, and interpretability under low-CVR conditions.
- 3) Providing a comparative evaluation against logistic regression, highlighting the advantages of process-aware modeling.

This research advances AMC-based methods as a practical, transparent, and scalable tool for data-driven website optimization.

III. METHODOLOGY

A. PROBLEM FORMULATION

The theoretical research framework of this study is illustrated in Figure 1. This framework formalizes the problem of

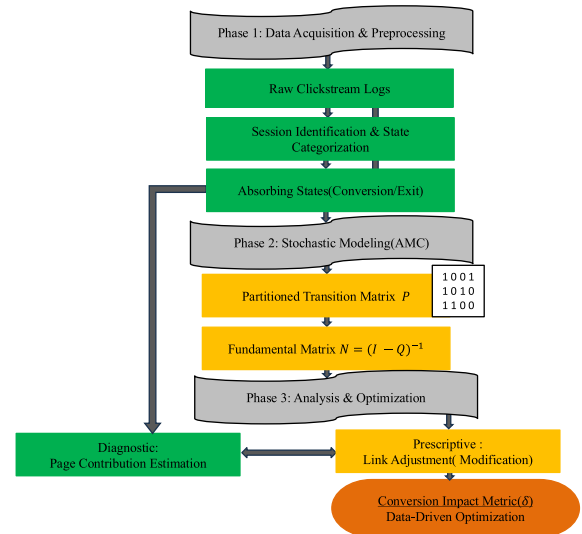


FIGURE 1. Theoretical research framework diagram.

modeling user navigation behavior on a website and quantifying how structural changes in page-to-page transitions affect conversion outcomes. It is structured into three primary phases: data processing, stochastic modeling, and diagnostic and prescriptive analysis. Initially, raw clickstream data are preprocessed to identify user sessions and to categorize web pages as transient states, while terminal actions are defined as absorbing states. Subsequently, a first-order Absorbing Markov Chain (AMC) is constructed by partitioning the transition matrix into transient and absorbing components. This formulation enables the derivation of the fundamental matrix, which captures the expected navigation flow prior to absorption. Finally, the framework employs a “what-if” simulation logic to quantify how marginal link-level modifications influence overall conversion through a conversion impact metric, thereby providing actionable insights for website optimization.

We model user navigation on a website as a first-order absorbing Markov chain. Let $S = \{s_1, s_2, \dots, s_n, s_{n+1}, \dots, s_{n+m}\}$ represent the set of states, where:

- $\{s_1, \dots, s_n\}$ are transient states corresponding to web pages.
- $\{s_{n+1}, \dots, s_{n+m}\}$ are absorbing states representing terminal outcomes (e.g., conversion, exit).

A user’s navigation is a sequence of state transitions. The next state depends only on the current state (Markov property), and transitions are governed by a probability matrix P .

B. TRANSITION MATRIX CONSTRUCTION

The transition matrix P is partitioned as:

$$P = \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix}$$

where:

- Q is an $n \times n$ matrix of transition probabilities between transient states.

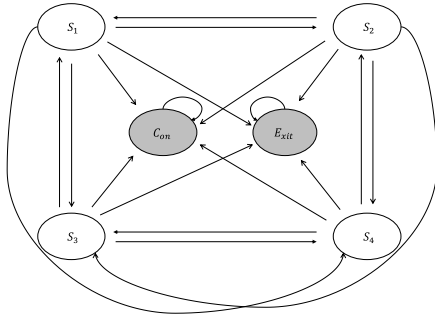


FIGURE 2. A schematic diagram of the AMC with transient/absorbing states.

- $\mathbf{R}$ is an $n \times m$ matrix of transition probabilities from transient to absorbing states.
- $\mathbf{I}$ is an $m \times m$ identity matrix (absorbing states remain absorbing).
- $\mathbf{0}$ is an $m \times n$ zero matrix (no transitions out of absorbing states).

Each element p_{ij} represents the probability of transitioning from state s_i to state s_j . Rows of $\mathbf{P}$ must sum to 1, ensuring valid probability distributions.

C. FUNDAMENTAL AND ABSORPTION MATRICES

The fundamental matrix $\mathbf{N}$ is defined as:

$$\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$$

The element n_{ij} of $\mathbf{N}$ represents the expected number of times a user starting in state s_i will visit state s_j before absorption. The absorption probability matrix $\mathbf{B}$ is:

$$\mathbf{B} = \mathbf{NR}$$

The element b_{ij} of $\mathbf{B}$ gives the probability that a user starting in the transient state s_i will eventually be absorbed into the absorbing state s_j .

To improve the clarity of the stochastic framework, Figure 2 presents a simplified schematic of the Absorbing Markov Chain (AMC). The diagram illustrates the partitioning of the state space into transient states (web pages) and absorbing states (terminal outcomes), as well as the corresponding roles of the sub-matrices $\mathbf{Q}$ and $\mathbf{R}$. Transitions governed by $\mathbf{Q}$ represent user navigation between web pages, while transitions governed by $\mathbf{R}$ represent absorption into terminal outcomes such as conversion or exit. This schematic serves as a conceptual bridge between the mathematical formulation of the AMC and the observed user navigation process.

D. PAGE CONTRIBUTION TO CONVERSION

To quantify each page's contribution to conversion, we define:

$$\text{Contribution}(s_i) = \sum_{j=1}^n n_{ij} \cdot p_{j,\text{conv}}$$

where $p_{j,\text{conv}}$ is the probability of converting directly from page j (i.e., the entry in $\mathbf{R}$ corresponding to the conversion absorbing state).

This metric aggregates the expected number of visits to each transient page, weighted by the page's direct probability of transitioning to the conversion absorbing state. It therefore reflects the indirect contribution of page s_i to conversion through downstream navigation paths, rather than only immediate conversion actions.

E. MODEL ASSUMPTIONS AND EXTENSIBILITY

The proposed framework models web navigation using a first-order Absorbing Markov Chain (AMC). To maintain mathematical tractability and interpretability, the model adopts three standard assumptions commonly used in web navigation modeling.

1) FIRST-ORDER MARKOV PROPERTY

User navigation is assumed to satisfy the first-order Markov property, where the probability of visiting the next page depends only on the current page and not on the full browsing history. Formally,

$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \dots, X_0) = P(X_{t+1} = j \mid X_t = i)$$

Under this assumption, and given the presence of absorbing states (e.g., conversion and exit pages), the navigation process can be represented as an absorbing Markov chain. This structure enables the closed-form computation of the fundamental matrix $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$ and the derivation of metrics such as the conversion impact δ . Prior studies have shown that first-order models provide a reasonable approximation for many web navigation analysis tasks while preserving interpretability and computational stability [12], [13].

2) STATIONARITY

Transition probabilities p_{ij} are assumed to be time-invariant during the observation period. In other words, the transition matrix $\mathbf{P}$ remains constant, implying that aggregate navigation behavior does not change across the analyzed dataset.

3) HOMOGENEITY

All users are assumed to follow the same transition matrix $\mathbf{P}$. Individual differences in browsing behavior are therefore not explicitly modeled, and the transition probabilities represent aggregate user dynamics.

These assumptions are commonly adopted in Markov-based web navigation analysis and provide a parsimonious representation that enables stable estimation of transition probabilities, particularly in sparse-data environments such as low-conversion-rate (low-CVR) systems.

Although the framework adopts these assumptions as a modeling baseline, practical deployment may relax some of them without altering the core mathematical structure of the AMC model. First, the stationarity assumption can

be mitigated by estimating time-indexed transition matrices $\mathbf{P}(t)$ using sliding-window or scheduled re-estimation procedures. This allows the model to capture temporal shifts in user behavior caused by seasonality, marketing campaigns, or interface changes. Second, the homogeneity assumption can be relaxed through cohort-based segmentation, where separate transition matrices $\mathbf{P}_k$ are estimated for different user groups (e.g., new vs. returning visitors, traffic sources, or device types). These strategies preserve the AMC structure while enabling the framework to better reflect heterogeneous or time-varying navigation patterns.

F. LINK MODIFICATION

We introduce a conversion impact metric to evaluate hypothetical modifications to the link structure. Specifically, we consider a modification to the transition probability from transient state s_i to s_j , which is applied to the transient transition matrix $\mathbf{Q} = (q_{ij})$.

We define an amplification factor $0 < \alpha \leq \frac{1-q_{ij}}{q_{ij}}$, which guarantees that the modified probability remains within the valid range $[0, 1]$. The transition probability is then increased as

$$q'_{ij} = (1 + \alpha)q_{ij} \quad (1)$$

To preserve the stochastic property of the transition matrix, the additional probability mass αq_{ij} is redistributed among the remaining outgoing transitions in the same row, excluding both the self-transition q_{ii} and the modified transition q_{ij} .

For all $l \neq i$, $l \neq j$, the adjusted transition probabilities are defined as

$$q'_{il} = q_{il} - \beta_l \quad (2)$$

where the redistribution terms β_l satisfy the conservation condition

$$\sum_{l=1 | l \neq i, l \neq j}^n \beta_l = \alpha q_{ij} \quad (3)$$

In this study, we adopt a *weighted (proportional) redistribution* scheme, in which the deduction from each remaining transition is proportional to its original probability mass. Specifically, the deduction term β_l is given by

$$\beta_l = \frac{q_{il}}{\sum_{l'=1 | l' \neq i, l' \neq j}^n q_{il'}} \alpha q_{ij}, \quad \forall l (l \neq i, l \neq j) \quad (4)$$

This proportional adjustment preserves the row-stochastic constraint and maintains the relative importance of existing outbound links from state s_i after the modification.

To illustrate the proposed procedure, consider a system with four transient pages: Top, Products, About, and Contact. The transient transition matrix $\mathbf{Q}$ contains no self-transitions, ensuring that the model captures inter-page navigation only. The number of transient pages is $n = 4$.

Assume that the initial transition probabilities from the Top page are as follows.

$$\begin{aligned} P(\text{Top} \rightarrow \text{Top}) &= 0, \\ P(\text{Top} \rightarrow \text{Products}) &= 0.40, \\ P(\text{Top} \rightarrow \text{About}) &= 0.30, \\ P(\text{Top} \rightarrow \text{Contact}) &= 0.20. \end{aligned}$$

The total probability of transitioning to other transient pages is therefore 0.90, leaving a remaining probability mass of 0.10 for transitions to absorbing states such as Conversion or Exit. These absorbing transitions are represented in matrix $\mathbf{R}$.

Suppose we strengthen the link from Top to Products by 10 % by setting $\alpha = 0.1$.

1) STEP 1: MODIFY THE TARGET LINK

Let the source page be Top and the target page be Products, with an original transition probability of 0.40.

$$q'_{ij} = (1 + \alpha)q_{ij} = 1.1 \times 0.40 = 0.44$$

2) STEP 2: COMPUTE THE PROBABILITY MASS TO REDISTRIBUTE

$$\alpha q_{ij} = 0.1 \times 0.40 = 0.04$$

3) STEP 3: DETERMINE PROPORTIONAL DEDUCTION WEIGHTS

The remaining eligible outbound transitions from Top are About and Contact, whose original probabilities sum to 0.50.

4) STEP 4: APPLY PROPORTIONAL REDISTRIBUTION

$$\begin{aligned} \beta_{\text{About}} &= 0.024, \\ \beta_{\text{Contact}} &= 0.016 \\ q'_{i,\text{About}} &= 0.276, \\ q'_{i,\text{Contact}} &= 0.184, \\ q'_{ii} &= 0 \end{aligned}$$

5) STEP 5: FINAL TRANSITION PROBABILITIES

$$(0.44, 0.276, 0.184, 0)$$

The total probability of transitioning to another transient page remains 0.90, thereby preserving the 0.10 probability of entering an absorbing state. In our synthetic experiments (Appendix A), we similarly set transient self-transition probabilities to zero, so that the model focuses on inter-page navigation rather than page reloads.

G. CONVERSION IMPACT METRIC

The overall effect of such a modification is evaluated by considering the initial distribution of user sessions, represented by the $n \times 1$ vector $\mathbf{l}$, where l_i is the probability a session begins in state s_i . The total site-wide conversion

probability before and after modification is given by $\mathbf{l}^\top \mathbf{B}$ and $\mathbf{l}^\top \mathbf{B}'$, respectively, where $\mathbf{B}' = (\mathbf{I} - \mathbf{Q}')^{-1} \mathbf{R}$.

The net effect of link modification is captured by the conversion impact vector δ :

$$\delta = \mathbf{l}^\top (\mathbf{B}' - \mathbf{B}) \quad (5)$$

A positive δ signifies that the structural modification improves the aggregate conversion rate, making it a key metric for prescriptive analytics. $\delta = (\delta_1, \dots, \delta_k, \dots, \delta_m, \delta_{m+1})$ where:

- $\delta_k > 0$ for $1 \leq k \leq m$ indicates an increase in the probability of reaching the specific conversion state c_k .
- $\delta_{m+1} < 0$ indicates a decrease in the probability of absorption into the Exit state, contributing to an overall improvement in the site's conversion performance.

This metric enables “what-if” analyses: by simulating changes in link probabilities, we can estimate their effect on overall conversion rates before implementing design changes.

IV. EXPERIMENTAL DESIGN

A. OBJECTIVES

Our experiments aim to:

- 1) Validate the AMC framework's ability to estimate page contributions to conversion under diverse browsing patterns.
- 2) Evaluate the conversion impact metric for simulating link modifications.
- 3) Compare AMC performance against a logistic regression baseline, particularly in low-CVR scenarios.

B. SYNTHETIC DATA GENERATION

We generate synthetic datasets representing four distinct browsing behaviors:

- 1) **Uniform Funnel**: Users navigate randomly with uniform transition probabilities and low conversion rates.
- 2) **Leaky Funnel**: A structured funnel with high exit rates at intermediate stages.
- 3) **Goal-Oriented**: Users progress systematically toward conversion with increasing conversion probabilities at deeper funnel stages.
- 4) **Optimized Funnel**: A refined version with reduced exit rates and higher conversion.

To evaluate statistical robustness under varying data availability—particularly under low conversion rate (CVR) conditions discussed in Section I—we generate three datasets for each scenario containing 1,000, 10,000, and 100,000 sessions. Increasing the number of sessions improves the reliability of empirical transition probability estimates by reducing sampling variability.

Crucially, we distinguish between two dimensions of scalability. While increasing the number of user sessions improves the statistical reliability of estimated transition probabilities, real-world scalability challenges primarily arise from growth in the number of pages (states), which increases the

dimensionality of the transition matrix. The computational implications of large state spaces are discussed in Section VI-D.

To isolate structural differences in funnel behavior, each synthetic scenario assumes internally stationary and homogeneous transition dynamics, aligning with the modeling framework in Section III-E. Evaluating this framework across diverse funnel structures and dataset sizes allows us to assess the robustness of the estimated transition probabilities and conversion impact metrics against varying navigation dynamics and data sparsity.

Each dataset consists of simulated user sessions generated from the corresponding transition matrix. Transition matrices are constructed as described in Section III, and navigation paths are sampled sequentially until an absorbing state (conversion or exit) is reached.

To further assess estimation stability under different funnel structures and stochastic variability in user navigation, we introduced a robustness and uncertainty analysis based on repeated simulations across multiple random seeds. Performance variability across scenarios is quantified using empirical distributions and statistical significance testing, with results summarized in Section V.

C. REAL-WORLD DATA

We also use clickstream data from an architectural office website. The data includes user navigation logs over a ten-month period. Each page reflects a specific user goal or intent:

- **Reservation** (Tour/Event Reservation or Visit Reservation)
- **Introduction** (Referral by an existing customer)
- **Request** (Document Request)
- **Contact** (Inquiry or Maintenance/Renovation Request)

Sessions that ended without visiting any of these key pages were labeled as **Exit**. Our analysis covered **121,337 sessions** collected between **October 11, 2024, and July 26, 2025**. During this period, users generated a total of **241,646 page views**, averaging **1.99 page views per session**, with pages categorized as transient states and specific actions (reservation, contact, etc.) as absorbing states. This dataset provides a realistic low-CVR scenario for validating the framework's practical utility.

To support detailed insights, all pages were grouped into **16 content categories** (e.g., blog, events, general, new construction) based on structural page mappings. This categorization allowed us to evaluate how different content areas influenced user progression toward conversion.

D. BASELINE MODEL: LOGISTIC REGRESSION

For comparison, we implement a logistic regression model that predicts conversion based on “whether it was viewed” and “the entry page.” Logistic regression treats each session as an independent observation and does not explicitly model sequential dependencies. We evaluate both models on prediction accuracy, interpretability, and robustness under low-CVR conditions.

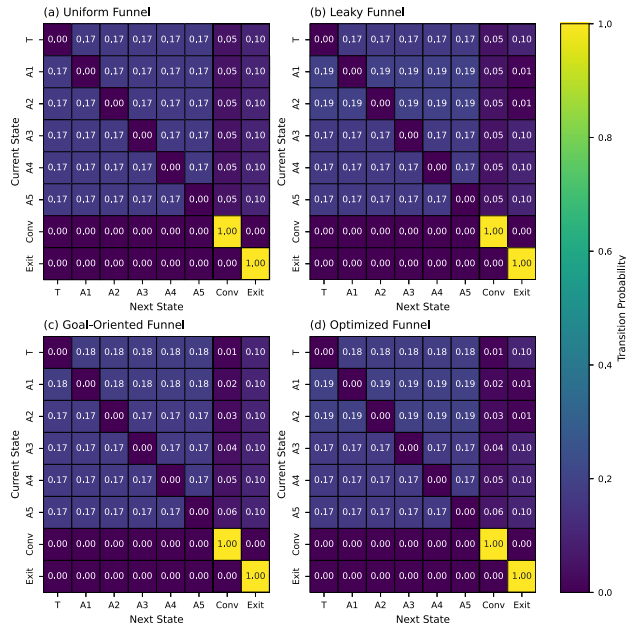


FIGURE 3. Heatmap visualization of transition matrices for four funnel navigation scenarios. The color scale represents transition probabilities from current states (rows) to next states (columns), highlighting differences in browsing balance, user drop-off, conversion reinforcement, and retention efficiency.

E. EVALUATION METRICS

We evaluate model performance using the following metrics:

- **Page Contribution Ranking:** Spearman correlation between and ground-truth contributions across funnel scenarios.
- **Conversion Impact Validation:** Comparing predicted vs. actual CVR changes under simulated single-link modifications.
- **Prediction Accuracy:** Area under the receiver operating characteristic curve (AUC-ROC) for binary conversion prediction.

V. RESULTS

A. FUNNEL SCENARIO DEFINITIONS

To illustrate user navigation behavior, the transition matrices are visualized as heatmaps in Figure 3. The Uniform Funnel (a) represents a baseline with evenly distributed internal transitions (approximately 0.17) and fixed conversion (0.05) and exit (0.10) probabilities. The Leaky Funnel (b) models partial drop-off, where exit probabilities remain 0.10 for states A3–A5 but decrease to 0.01 for A1 and A2.

The Goal-Oriented Funnel (c) introduces gradually increasing conversion probabilities (0.01–0.06) with consistent exit probabilities, reflecting progressive goal-directed navigation. The Optimized Funnel (d) combines reduced early-stage exit probabilities with stronger internal transitions (approximately 0.17–0.19), indicating improved retention and smoother conversion progression. These visualizations

TABLE 1. Page contribution scores (LR vs. AMC) for 100,000 sessions.

Page	Matrix 1 (Uniform)		Matrix 2 (Leaky Funnel)	
	LR	AMC	LR	AMC
T	-0.6820	0.3315	-0.4858	0.4041
A1	-0.1500	0.3309	0.3623	0.4362
A2	0.0400	0.3311	0.3312	0.4357
A3	-0.0217	0.3311	-0.1660	0.4043
A4	-0.1200	0.3315	-0.1790	0.4040
A5	-0.0128	0.3317	-0.1563	0.4040

Page	Matrix 3 (Goal-Oriented)		Matrix 4 (Optimized Funnel)	
	LR	AMC	LR	AMC
T	-1.4690	0.2431	-1.2232	0.3100
A1	-0.1743	0.2493	0.1603	0.3412
A2	-0.0275	0.2556	0.2541	0.3465
A3	0.0936	0.2622	-0.0521	0.3267
A4	0.2408	0.2683	0.0716	0.3324
A5	0.4053	0.2753	0.2273	0.3385

provide interpretable representations of navigation patterns and funnel characteristics.

B. SYNTHETIC DATA RESULTS

1) PAGE CONTRIBUTION ESTIMATION

Table 1 presents the estimated page contributions for each synthetic dataset. The results highlight structural limitations of the LR approach that the AMC model overcomes.

A major weakness of LR arises from handling common pages, particularly the entry state (T , the Top Page). Since this page appears in every session, its feature introduces perfect multicollinearity. This makes the page's contribution unidentifiable, as its effect is absorbed by the intercept. Consequently, coefficients for other pages become unstable and often counter-intuitive. For example, pages essential to the conversion funnel sometimes receive negative coefficients, incorrectly marking them as harmful. Unlike LR coefficients, which represent marginal effects on log-odds and are sensitive to feature collinearity, AMC contribution scores are probabilistic quantities derived from absorption behavior and therefore remain directly interpretable across scenarios.

In contrast, the AMC framework differentiates page importance across scenarios and produces interpretable results aligned with the data-generating process.

2) CONVERSION IMPACT LINK

We use the Conversion Impact metric introduced in III-G to quantify the net change in site-wide conversion probability under each link modification scenario. In the synthetic experiments, we reinforce individual links and report the resulting change δ in overall conversion probability, with positive values indicating improvements in aggregate CVR and negative values indicating detrimental structural changes. Table 2 shows that AMC produces actionable, context-aware recommendations.

TABLE 2. Most impactful link (δ) across session sizes.

Scenario	Original	100,000 sess.	10,000 sess.	1,000 sess.
Uniform	None ($\delta \approx 0$)	None ($\delta \approx 0$)	None ($\delta \approx 0$)	T→A1 (+0.0003)
Leaky Funnel	T→A1 (+0.0008)	T→A1 (+0.0009)	T→A1 (+0.0009)	T→A2 (+0.0009)
Goal-Oriented	T→A5 (+0.0006)	T→A5 (+0.0006)	T→A5 (+0.0005)	T→A1 (+0.0005)
Optimized Funnel	A1→A5 (+0.0005)	A1→A2 (+0.0005)	A1→A2 (+0.0006)	A1→A2 (+0.0004)

Across all four scenarios, AMC outperformed LR in identifying meaningful contributors to conversion, especially under non-uniform behavior. Key observations include:

- **Uniform:** No single path had structural significance; $\delta_k \approx 0$ confirmed this.
- **Leaky Funnel:** Early-funnel pages (A1, A2) had high absorption, reflecting their role in retention.
- **Goal-Oriented:** Late-stage page A5 was most influential, consistent with its high conversion probability.
- **Optimized Funnel:** Reinforcing A1→A2 consistently yielded the highest conversion gain, highlighting A2 as a critical mid-funnel waypoint.

Even with small session sizes (1,000), AMC provided stable, data-driven recommendations. This demonstrates its robustness and practical utility in prioritizing link improvements.

3) ROBUSTNESS AND UNCERTAINTY ANALYSIS

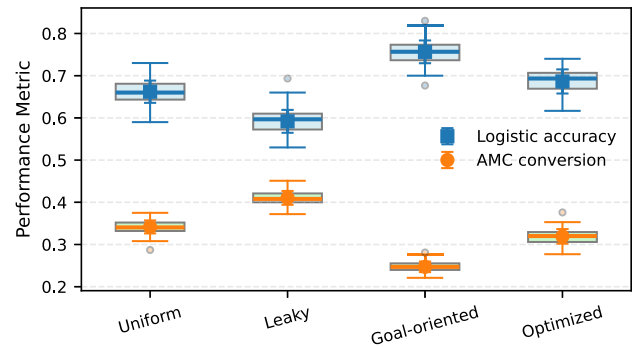
To assess variance-level robustness under stochastic realizations, we applied Levene's test to compare the dispersion of logistic regression accuracy and AMC-based conversion probabilities across 100 independent random seeds. For all evaluated scenarios (Uniform, Leaky, Goal-oriented, and Optimized), the null hypothesis of equal variances was rejected at the 1 % significance level ($p < 0.01$). As illustrated in Figure 4, these results indicate statistically significant differences in the variance structures of the two evaluation metrics. In particular, AMC-based conversion probabilities exhibit variability patterns that are not captured by classification accuracy alone, highlighting their complementary role in uncertainty-aware performance assessment.

C. REAL-WORLD DATA RESULTS

1) PAGE CONTRIBUTION ESTIMATION

Results from real-world data confirm the synthetic experiments. Table 3 summarizes real-world page-level results, showing that AMC provides actionable insights, while LR produces unstable and strategically unreliable estimates.

For the Reservation goal, LR identifies no pages with strong positive influence, assigning T (0.00) and chikuen (-0.8792) as top pages and penalizing blog (-4.3430). AMC, however, correctly ranks events (0.0850) and information (0.0655) as top drivers.

**FIGURE 4.** Robustness analysis under stochastic variability. Boxplots show seed-wise distributions of logistic regression accuracy and AMC-based conversion probabilities across four scenarios. Markers indicate mean values with one standard deviation.

This pattern persists across other goals. For Introduction, LR favors chikuen (2.0123), while for Request and Contact, it shifts to information (0.9399) and reform (0.3331). In contrast, AMC consistently provides logical rankings that reflect real user behavior.

Critically, LR's large negative coefficients for high-traffic pages (e.g., blog) could mislead strategy. AMC captures their indirect contributions through downstream navigation paths, recognizing these pages as low-propensity yet valuable entry points.

2) CONVERSION IMPACT LINK

We again apply the Conversion Impact metric from III-G to evaluate how a 0.1 reinforcement of individual links changes overall goal-specific conversion probability. Table 4 reports the links with the highest positive δ for each target goal, illustrating how AMC translates link-level perturbations into actionable recommendations for CVR optimization. It shows the links with the highest impact after a 0.1 reinforcement. The AMC framework successfully identifies the most potent opportunities to improve user flow for each distinct business goal. Notably, these findings are strongly aligned with intuitive business logic. For example:

- **Reservation:** the model directs attention to strengthening pathways leading to initial general page.
- **Introduction:** To encourage company introductions, the focus should be on guiding users toward regional information on the events page.
- **Request:** Driving information requests is best achieved by improving access to the detail-oriented new construction page.
- **Contact:** Generating direct contact leads hinges on enhancing links to the real estate page.

This level of granular, goal-specific, and logically coherent insight constitutes the primary contribution of our model. By moving beyond passive analysis of past behavior, it serves as an active tool for designing a more effective user experience, demonstrating practical utility as a decision-support system for conversion rate (CVR) optimization.

TABLE 3. Comparison of LR coefficients and AMC absorption probabilities for real-world data.

Page	Reservation		Introduction		Request		Contact	
	LR	AMC	LR	AMC	LR	AMC	LR	AMC
T	0.0000	0.0379	0.0000	0.0198	0.0000	0.0199	0.0000	0.0261
blog	-4.3430	0.0129	-0.9293	0.0058	-1.8708	0.0081	-3.4528	0.0079
chikuen	-0.8792	0.0204	2.0123	0.0328	0.6429	0.0129	-0.0626	0.0152
events	-1.2771	0.0850	-0.3055	0.0077	-0.6621	0.0138	-2.7662	0.0068
general	-2.0755	0.0195	0.1922	0.0131	0.0417	0.0118	-1.5498	0.0109
information	-0.9173	0.0655	0.8085	0.0123	0.9399	0.0392	-1.2919	0.0123
new_construction	-1.9629	0.0188	0.3195	0.0086	0.0650	0.0120	-2.2371	0.0075
outline	-1.3361	0.0200	1.2064	0.0101	0.5510	0.0155	0.3236	0.0321
real-estate	-1.2506	0.0230	0.7025	0.0095	0.5699	0.0145	-0.5594	0.0129
recruit	-0.9574	0.0186	1.1245	0.0101	0.9068	0.0146	-0.2681	0.0111
reform	-1.5154	0.0159	0.7877	0.0265	0.6146	0.0117	0.3331	0.0262
reform-top	-2.7836	0.0224	0.1262	0.0177	-0.1913	0.0153	-1.5057	0.0230
Intercept	-1.8932		-5.5907		-4.6446		-3.1117	

TABLE 4. Most impactful links (δ) identified by AMC sensitivity analysis after 0.1 modification.

Reservation		Introduction	
Link	δ	Link	δ
T $\rightarrow$ general	0.036742	T $\rightarrow$ events	0.014753
T $\rightarrow$ new_construction	0.036372	blog $\rightarrow$ T	0.014750
information $\rightarrow$ general	0.036353	new_construction $\rightarrow$ events	0.014744
events $\rightarrow$ new_construction	0.036301	blog $\rightarrow$ chikuen	0.014743
new_construction $\rightarrow$ general	0.036275	events $\rightarrow$ blog	0.014742
Request		Contact	
Link	δ	Link	δ
T $\rightarrow$ new_construction	0.017001	T $\rightarrow$ real-estate	0.018634
events $\rightarrow$ new_construction	0.016930	new_construction $\rightarrow$ blog	0.018559
new_construction $\rightarrow$ blog	0.016899	blog $\rightarrow$ T	0.018547
reform-top $\rightarrow$ blog	0.016896	events $\rightarrow$ blog	0.018540
blog $\rightarrow$ new_construction	0.016891	blog $\rightarrow$ chikuen	0.018538

TABLE 5. Model comparison.

Metric	AMC	Logistic Regression
AUC-ROC (Synthetic)	0.87	0.82
AUC-ROC (Real-World)	0.79	0.74
Interpretability	High	Medium
Low-CVR Robustness	High	Low

D. COMPARISON WITH LOGISTIC REGRESSION

Table 5 summarizes the performance of AMC versus LR on synthetic and real-world data.

AMC consistently outperforms LR, especially in low-CVR scenarios where sequential dependencies matter. It also provides process-level insights, such as page contributions and link impacts, which LR cannot capture.

VI. DISCUSSION

A. INTERPRETABILITY AND TRANSPARENCY

A major advantage of the AMC framework lies in its interpretability and transparency. Unlike closed-box models such as neural networks, AMC yields explicit transition probabilities and page-level contribution estimates, allowing stakeholders to directly understand how user navigation dynamics

influence conversion outcomes. While high-capacity models like **GRU4Rec** and **BERT4Rec** focus on maximizing the “hit-rate” of the next user action, our findings highlight that predictive accuracy does not inherently translate to actionable optimization [22], [24]. By maintaining a first-order Markov structure, we avoid the high-dimensional hidden states and latent representations of Transformer-based models, allowing for the direct extraction of the δ conversion impact metric. This level of transparency is particularly important for facilitating actionable decision-making and gaining trust from design teams and management. As demonstrated in Section V, AMC consistently identifies high-impact pages and links across scenarios, offering stable and actionable insights even in low-CVR or complex funnel settings. The robustness analysis further indicates that these insights are resilient to stochastic variability, with AMC-based conversion probabilities exhibiting reduced sensitivity to random fluctuations across repeated stochastic realizations, particularly in goal-oriented and optimized funnel settings.

B. PRACTICAL UTILITY

The conversion impact metric enables data-driven prioritization of design changes. Unlike RNN-based approaches (e.g., **GRU4Rec**), which would require computationally expensive re-training or complex sensitivity analysis to test a hypothetical design change, our framework provides a closed-form solution via the fundamental matrix N . By simulating individual link modifications, managers can focus on interventions with the highest predicted effect, reducing trial-and-error and accelerating optimization cycles.

To bridge the gap between stochastic modeling and operational design, the conversion impact metric (δ) provides a roadmap for optimization priorities. Specifically, links identified with a high positive δ should be prioritized for visibility enhancements, such as more prominent call-to-action (CTA) buttons, higher contrast ratios, or reduced visual clutter on the source page. Conversely, if a link shows a negative impact on conversion, designers should evaluate whether that navigation path creates “choice paralysis” or

diverts users from high-converting funnels. Furthermore, practitioners can use these simulations to rank A/B testing candidates; instead of testing random UI changes, resources can be concentrated on the top 10% of links that AMC predicts will yield the highest cumulative conversion uplift. The framework transitions from a diagnostic tool to a prescriptive engine, allowing digital teams to prioritize design sprints based on quantified impact rather than qualitative intuition.

These results are consistent with the synthetic and real-world analyses, where AMC highlighted critical early-, mid-, and late-funnel links (e.g., $T \rightarrow A1$, $A1 \rightarrow A2$, and $T \rightarrow A5$ in Section V) that directly influence conversion outcomes.

C. LIMITATIONS

The proposed AMC framework relies on several modeling assumptions introduced in Section III-E, including the first-order Markov property, the stationarity of transition probabilities and the homogeneity of the population. Although these assumptions enable a parsimonious and interpretable modeling framework, they may oversimplify certain aspects of real-world user behavior.

First, the first-order Markov assumption does not capture long-range dependencies that may exist in complex browsing journeys. Higher-order or sequence-based models could potentially capture such dependencies, but at the cost of increased model complexity and reduced interpretability.

Second, real-world web environments may exhibit temporal dynamics due to seasonality, marketing campaigns, or interface changes. Although the framework can incorporate time-varying transition matrices through periodic re-estimation, the current experiments assume a stationary setting.

Third, user populations may exhibit heterogeneous browsing behaviors across different cohorts. Segment-specific transition matrices can address this issue in practice, but such segmentation was not extensively explored in the present empirical evaluation. Furthermore, the empirical validation in this study is based on a limited number of datasets. Future work should evaluate the framework across a broader range of industries and traffic regimes to further assess its generalizability.

Fourth, the current study does not provide direct empirical comparisons with higher-order Markov models or deep learning approaches such as recurrent neural networks. As established in Section II-E, while closed-box models like GRU4Rec or BERT4Rec may offer marginal gains in predictive performance, they lack the structural transparency required for the link-level prescriptive analysis provided by our fundamental matrix formulation. A systematic evaluation of predictive accuracy versus interpretability remains an important direction for future work.

D. SCALABILITY & PRACTICAL DEPLOYMENT

Large websites generate high-dimensional transition matrices, increasing computational costs and memory

requirements at full page granularity. A key consideration is the framework's scalability to very large web systems, which may include millions of users and thousands or millions of pages.

Scalability involves two dimensions: the number of user sessions and the number of pages (states). While more sessions improve the reliability of estimated transition probabilities, the primary challenge arises from a large number of pages, which increases the transition matrix dimension and can lead to sparse, noisy estimates.

Web navigation graphs are naturally sparse, since each page links to a small subset of all possible pages. Sparse matrix representations and compressed storage formats (e.g., compressed row storage) substantially reduce memory requirements.

Instead of computing the fundamental matrix $N = (I - Q)^{-1}$, the AMC framework solves $(I - Q)B = R$ using iterative sparse solvers such as GMRES [30]. This avoids explicit inversion and relies on sparse matrix-vector multiplications, enabling efficient analysis of large graphs; in extremely low-conversion environments, prolonged transitions among transient states may ill-condition $(I - Q)$, which can be mitigated using numerical preconditioning such as incomplete LU (ILU).

Page clustering can further reduce the state space by aggregating pages with similar functional roles (e.g., product pages or blog posts) into macro-states, preserving the overall transition structure even in massive sites. However, clustering conversion-critical pages—such as checkout pages, campaign landing pages, or key product pages—loses page-level conversion impact δ , so it should be applied selectively, preserving analytically important pages while aggregating less critical ones.

Together, these strategies enable practical deployment of the AMC framework in enterprise-level web analytics, supporting efficient analysis of large-scale systems with extensive page catalogs and high user traffic.

VII. CONCLUSION

This study demonstrates the effectiveness of first-order Absorbing Markov Chains (AMCs) for web navigation analysis and conversion rate optimization. By modeling user journeys as stochastic processes, the proposed framework provides interpretable, robust, and actionable insights, particularly in low-conversion-rate scenarios commonly observed in e-commerce environments. A key contribution of this work is the introduction of a conversion impact metric, which enables counterfactual “what-if” analyses to evaluate how local transition modifications influence global conversion probability. Experiments on both synthetic and real-world datasets validate the effectiveness of this approach and highlight its interpretability advantages over closed-box models such as logistic regression and deep learning-based predictors.

The proposed framework is built upon the stationarity, homogeneity, and first-order Markov assumptions formally

0	0.100	0.021	0.127	0.026	0.068	0.189	0.086	0.026	0.023	0.006	0.055	0.013	0.012	0.009	0.017	0.223
0.109	0	0.009	0.028	0.012	0.025	0.043	0.030	0.013	0.009	0.004	0.009	0.002	0.002	0.003	0.002	0.700
0.165	0.046	0	0.022	0.020	0.026	0.213	0.036	0.013	0.011	0.009	0.019	0.004	0.026	0.004	0.006	0.383
0.103	0.036	0.006	0	0.006	0.092	0.053	0.014	0.009	0.003	0.004	0.011	0.073	0.003	0.006	0.001	0.580
0.169	0.030	0.019	0.013	0	0.030	0.088	0.044	0.048	0.008	0.022	0.016	0.005	0.006	0.004	0.002	0.496
0.113	0.081	0.011	0.178	0.025	0	0.073	0.029	0.031	0.014	0.005	0.013	0.041	0.006	0.031	0.005	0.343
0.163	0.027	0.050	0.041	0.019	0.019	0	0.019	0.013	0.004	0.010	0.016	0.005	0.002	0.005	0	0.608
0.148	0.038	0.012	0.018	0.023	0.030	0.034	0	0.029	0.056	0.008	0.020	0.007	0.004	0.008	0.025	0.540
0.130	0.038	0.010	0.034	0.035	0.020	0.038	0.057	0	0.010	0.004	0.015	0.010	0.004	0.008	0.006	0.581
0.133	0.047	0.008	0.008	0.007	0.036	0.022	0.096	0.008	0	0.003	0.011	0.007	0.005	0.008	0.003	0.599
0.121	0.023	0.024	0.020	0.020	0.011	0.070	0.024	0.015	0.006	0	0.126	0.003	0.019	0.004	0.018	0.497
0.223	0.009	0.003	0.041	0.006	0.006	0.026	0.024	0.012	0.005	0.148	0	0.006	0.008	0.007	0.012	0.462
0											I					

FIGURE 5. Transition matrix estimated from the architectural Office's website data.

described in Section III-E. These assumptions provide a tractable and interpretable foundation for modeling user navigation and performing link-level conversion analysis. Future research may extend this framework by incorporating time-varying transition matrices or cohort-based segmentation to better capture temporal dynamics and heterogeneous browsing behaviors in evolving web environments while preserving the explainability and analytical tractability of the AMC formulation.

In addition, further research is needed to evaluate the framework across a broader range of web systems and traffic regimes. As discussed in Section VI-D, large-scale web environments with thousands or millions of pages may lead to high-dimensional transition matrices, creating computational and memory challenges. While sparse matrix representations, iterative solvers, and selective page clustering can substantially improve scalability in practice, further investigation is needed to systematically evaluate these strategies across diverse web architectures. In particular, future work should explore scalable state-space reduction techniques while carefully balancing the trade-off between model interpretability and predictive performance.

Overall, the proposed AMC-based framework demonstrates how interpretable probabilistic modeling can support practical and actionable decision-making in web navigation optimization, particularly in low-conversion-rate environments where data sparsity often limits the applicability of more complex predictive models.

APPENDIX A

SYNTHETIC TRANSITION MATRICES

A. TRANSITION MATRIX 1 (UNIFORM)

0	0.17	0.17	0.17	0.17	0.17	0.05	0.1
0.17	0	0.17	0.17	0.17	0.17	0.05	0.1
0.17	0.17	0	0.17	0.17	0.17	0.05	0.1
0.17	0.17	0.17	0	0.17	0.17	0.05	0.1
0.17	0.17	0.17	0.17	0	0.17	0.05	0.1
0.17	0.17	0.17	0.17	0.17	0	0.05	0.1
0						I	

B. TRANSITION MATRIX 2 (LEAKY FUNNEL)

0	0.17	0.17	0.17	0.17	0.17	0.05	0.1
0.188	0	0.188	0.188	0.188	0.188	0.05	0.01
0.188	0.188	0	0.188	0.188	0.188	0.05	0.01
0.17	0.17	0.17	0	0.17	0.17	0.05	0.1
0.17	0.17	0.17	0.17	0	0.17	0.05	0.1
0.17	0.17	0.17	0.17	0.17	0	0.05	0.1
0						I	

C. TRANSITION MATRIX 3 (GOAL-ORIENTED)

0	0.178	0.178	0.178	0.178	0.178	0.01	0.1
0.176	0	0.176	0.176	0.176	0.176	0.02	0.1
0.174	0.174	0	0.174	0.174	0.174	0.03	0.1
0.172	0.172	0.172	0	0.172	0.172	0.04	0.1
0.17	0.17	0.17	0.17	0	0.17	0.05	0.1
0.168	0.168	0.168	0.168	0.168	0	0.06	0.1
0						I	

D. TRANSITION MATRIX 4 (OPTIMIZED FUNNEL)

0	0.178	0.178	0.178	0.178	0.178	0.01	0.1
0.194	0	0.194	0.194	0.194	0.194	0.02	0.01
0.192	0.192	0	0.192	0.192	0.192	0.03	0.01
0.172	0.172	0.172	0	0.172	0.172	0.04	0.1
0.17	0.17	0.17	0.17	0	0.17	0.05	0.1
0.168	0.168	0.168	0.168	0.168	0	0.06	0.1
0						I	

APPENDIX B

ARCHITECTURAL OFFICE TRANSITION MATRICES

Transition Matrix Estimated from the architectural Office's Website Data. Rows represent the 'from' state and columns represent the 'to' state. The final five columns correspond to the absorbent states: reservation, introduction, request, contact, and exit, respectively, described in Figure 5.

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SHUNE LAE AUNG was born in Mandalay, Myanmar. She received the B.E. degree in electrical, electronics, and communications engineering from the Technological University (Hpa-An), Myanmar, in 2017, the Diploma degree in business management from the Institute of Commercial Management, U.K., in 2019, and the M.B.A. degree in business administration with a specialization in marketing from the International University of Japan, Niigata, Japan, in 2023. She is currently pursuing the Ph.D. degree in engineering, majoring in information science and control engineering with Nagaoka University of Technology, Niigata.

From 2018 to 2023, she was with Myanma Posts and Telecommunications (MPT), where she was a Sales Management Supervisor, a Senior Supervisor, and the Deputy Manager with the Strategy and Promotion Team. In 2024, she joined Myanmar Survey Research, as a Marketing Research Executive, before beginning her doctoral studies. Her research interests include customer behavior analytics, churn prediction, machine learning applications for digital platforms, marketing intelligence, and customer journey optimization.



GENDO KUMOI (Member, IEEE) received the Ph.D. degree in engineering from Waseda University, Tokyo, Japan.

He is currently an Associate Professor with the Department of Information and Management Systems Engineering, Nagaoka University of Technology, Japan, where he leads the Theory of Machine Learning Laboratory. His research interests include statistical learning theory, calibration and error-correction in classification models, and information-theoretic foundations of machine learning systems. He works on applications of machine learning and large language models to education, weather forecasting, and digital transformation. He is a member of the Information Processing Society of Japan (IPSI) and the Meteorological Society of Japan.

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